Alexander Friedrichsen

Applied AI Engineer

Burlington, VT, US | 1-802-922-8588 | alex.p.friedrichsen@gmail.com

ABOUT

Applied AI engineer who brings competitive intensity to production systems. World Champion card game player turned AI builder. Deployed a fine-tuned GPT vision pipeline projected to save $3M+ annually; designed a 12-agent orchestration platform with autoresearch, model routing, RAG-based retrieval, and automated quality gates. My skillset specializes in making correct generalizations from insufficient data. Published ML researcher (GECCO '23), full-stack Python/TypeScript/cloud. Open to relocation (NYC, SF).

WORK EXPERIENCE

Data Scientist, Union Mutual
May, 2025 - Present
AI lead at a mutual insurance company. Built production LLM systems and drove organization-wide AI adoption.

- Co-designed and deployed an AI inspection evaluation pipeline — fine-tuned GPT vision model processes property images against underwriter-designed rubrics, outputs structured risk assessments, and generates underwriter reports — projected to save $3M+ annually. Runs in automated batch mode across policies and on-demand per underwriter request
- Restructured production codebases for LLM agent integration; established CI/CD, version control (GitHub), and isolated dev/prod environments
- Led company-wide AI adoption: trained analytics team on agentic coding tools (Cursor, custom agents, MCP hooks), delivered company-wide lunch-and-learn demos, and represented the company at an agentic AI conference
- Conduct exploratory modeling, feature engineering, and SQL querying across AWS (App Runner, Redshift, S3) to support business decision-making

Pro Trading Card Game Competitor, Team Handshake Moxfield
Oct, 2023 - Present

- Won the Wonders of the First World Championship; Top 8 at Pro Tour Karlov Manor (Magic: The Gathering, $500K prize pool events)
- Built MTG Meta Analyzer — full-stack web app (JavaScript, Supabase, PostgreSQL) analyzing 20,000+ competitive match outcomes with ELO rating system, deployed at honestafblog.com
- Designs and leads testing process for 16-hour/day, 7-day testing sprints ahead of major tournament events
- Implemented Openclaw bot for automated organization and data gathering across team testing process
- Brokered team sponsorships and coordinated international travel logistics for 15-person competitive roster

Software Engineer II, BAE Systems
Aug, 2023 - May, 2024 9 months

- Built multi-source Power BI data models with DAX aggregations, delivering weekly dashboards that drove labor and product line decisions for senior stakeholders
- Architected internal staffing solutions using SharePoint and MS Access, serving 16,000 employees
- Spearheaded ETL pipeline integrating financial data, enabling transition from manual Excel analysis to predictive analytics

Software Developer, Social Ecological Gaming and Simulation Lab
Aug, 2022 - Jul, 2023 11 months

Public Health Analyst I, Vermont Department of Health
Feb, 2019 - Sep, 2021 2 years 7 months

PROJECTS EXPERIENCE

Multi-Agent AI Orchestration System
- Present
Production multi-agent system with 12 specialized AI agents coordinated through an Obsidian-backed knowledge vault. Self-improving via nightly research pipelines that mine X, Hacker News, and Reddit for agent-tooling advances and feed them back into the harness. Deterministic verification gates via hooks, parallel worktree isolation for concurrent code generation, and hybrid retrieval (BM25, dense vector, HyDE) over a 300-plus document knowledge base.

SKILLS

AI Systems
LLM Integration | Agentic Workflows | RAG | Structured Extraction | LLM Fine-Tuning | Prompt Engineering | Computer Vision | Agent Orchestration | MCP (Model Context Protocol)

Engineering
Python | TypeScript | JavaScript | SQL | Full-Stack Development | Docker | CI/CD | Git/GitHub | Unit Testing

Cloud & Data
AWS (App Runner, Redshift, S3) | PostgreSQL | Supabase | Google BigQuery | Cloud Infrastructure | ETL Pipelines

ML & Analytics
Statistical Modeling | Scikit-learn | A/B Testing | Causal Inference | R | Power BI (DAX)

EDUCATION

Data Science and Complex Systems, M.S., University of Vermont, Complex Systems Center
Aug, 2021 - May, 2023

Data Science, B.S., University of Vermont, College of Engineering and Mathematics
Aug, 2018 - May, 2022
Minor: Economics | Minor: Mathematics | Minor: Computer Science | Minor: Statistics

PUBLICATIONS

Evolutionary Machine Learning for Robust Facility Placements, GECCO '23 (Genetic and Evolutionary Computation Conference)
Published on: Jul 01, 2023
Published research on evolutionary algorithms for optimization under uncertainty. Also presented at JuliaCon.